

Yohan Vergis Vinu

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EDUCATION

University of Pennsylvania

Philadelphia, PA

Master of Science in Engineering, Computer and Information Science (GPA: 3.90/4)

Aug. 2025 – May 2027

- **Coursework:** Distributed Systems, Analysis of Algorithms, ML, Machine Perception, Big Data Analytics

SRM Institute of Science and Technology

Chennai, India

Bachelor of Technology, Computer Science and Engineering (GPA: 9.80/10, Top 1%)

Aug. 2021 – May 2025

EXPERIENCE

Graduate Teaching Assistant - CIS 5210 (Artificial Intelligence)

Jan. 2026 – Present

University of Pennsylvania

Philadelphia, PA

- Hold weekly office hours and debug sessions for **250+ students** in a graduate AI course covering search, MDPs, and probabilistic reasoning; assist with assignment design and grading across exam cycles

Software Engineer Intern, Applied AI

Aug. 2024 – June 2025

Moative

Chennai, India

- Eliminated manual recruiter screening across **75+ live interviews** by building **Vina**, an agentic CHRO platform that automated rubric generation, analysis, and candidate ranking (Django, LangGraph, PostgreSQL, AWS)
- Cut permit issuance time from **3–4 weeks to minutes** for Rowlett, TX by building an **AI permitting agent** that automated document verification and audit trail generation (Django, LangGraph, AWS)
- Replaced manual call center workflows for **30+ concurrent users** by designing a **multi-channel utility agent** handling outage and billing support across web, voice, and SMS (Flask, PostgreSQL, ElevenLabs, Twilio)
- Reduced company firmographic lookup to **seconds** with **95% NAICS classification accuracy** by building an async scraping and AI agent pipeline exposed via a REST API (FastAPI, LangChain, Playwright)

Machine Learning Engineer Intern

Aug. 2023 – Mar. 2024

Samsung Research

Chennai, India

- Improved multi-class pet detection accuracy by **7% over baseline** by fine-tuning and benchmarking **ResNet**, **YOLOv8**, and **MobileNetV2-SSD** across standardized evaluation sets
- Compiled and annotated a **10K+ image dataset** through manual collection, labeling, and augmentation to improve class balance, earning **Samsung's Certificate of Excellence**

PROJECTS

LLMFuzz: Distributed LLM-Guided Fuzzing Agent | Claude SDK, Redis Streams, coverage.py

Mar. 2026

- Outperformed random fuzzing by **7.6 pp** (98.3% vs 90.7% branch coverage) using **53% fewer inputs**, finding **4 crashes** a random baseline missed via a coverage-guided ReAct loop with adaptive strategy selection
- Scaled fuzzing across stateless workers with zero-config horizontal scaling by building a Redis Streams consumer group architecture; coordinator auto-reassigns low-coverage targets with alternative strategies

Pantry.ai: AI-Native Kitchen OS | Next.js, FastAPI, Claude API, Redis Streams, PostgreSQL

Jan. 2026

- Built a full-stack AI kitchen assistant with **persistent state** and **multimodal inventory updates** – Claude extracts ingredients from photos, Whisper handles voice, and text input updates stock in real time across sessions
- Designed a **distributed pipeline** using Redis Streams – parallel workers handle web search, LLM generation, macro optimization, and expiry prioritization concurrently, with a coordinator merging and ranking results

PennSearch: Distributed Cloud Search Engine | Java, AWS EC2, Distributed Systems

Dec. 2025

- Achieved **sub-50ms** query latency across **300k+ indexed pages** by engineering a crawl-index-rank pipeline with multi-signal ranking (TF-IDF, cosine similarity, PageRank), robots compliance, and language filtering
- Ensured fault tolerance and horizontal scalability by building a distributed search engine with a custom web server, distributed key-value store, and parallel analytics engine on AWS EC2, designed for node failure recovery

TECHNICAL SKILLS

Programming Languages: Python, SQL, Java, TypeScript, JavaScript, C/C++

AI/ML: LangGraph/LangChain, LangSmith, NumPy, Pandas, PyTorch, Hugging Face, TensorFlow, OpenCV

Web Dev. & APIs: Django, FastAPI, Flask, React, Next.js, Streamlit, REST APIs, Tailwind CSS, Postman

Cloud, Databases & DevOps: AWS, PostgreSQL, Docker, Git/GitHub, GitHub Actions, ChromaDB, GCP, Redis